

Payam Parvizi

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Professional Summary

PhD-trained Research Associate with 4+ years of experience developing and validating reinforcement learning methods for continuous control under real-world-motivated constraints. I am driven by a deep curiosity about how learning-based controllers can achieve the smooth, stable, and reliable behavior needed for safe deployment in physical systems. I specialize in action/policy regularization and policy smoothing to reduce high-frequency oscillations and improve reliability, validated through the development of custom Gymnasium-compatible environments and simulation-to-real transfer. My work includes deployment on quadcopter platforms and applications in satellite-to-ground optical communications, conducted at the University of Ottawa in collaboration with the National Research Council of Canada (NRC). Also, contributed over 2 years to an R&D project on robotic manipulation, focused on transferring human motion skills to a robot in a TÜBİTAK-funded project.

Core Skills

AI & Machine Learning	Reinforcement Learning (PPO, SAC, TD3, DDPG), Supervised Learning, Custom Simulation Environment Design, Regularization Techniques, Reward Shaping, Robustness & Stability Evaluation
Deep Learning	Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs)
Frameworks & Libraries	PyTorch, TensorFlow, JAX (XLA GPU backend), MuJoCo Playground, Gymnasium, DM-Control, Tianshou, W&B
Programming & Scripting	Python, MATLAB, Simulink, Bash, Arduino
Tools	Git, LaTeX, Ubuntu/Linux, HPC/SLURM (Compute Canada)
Control Systems	System Dynamics; Classical Control design (PID, lead/lag); Modeling in Simulink
CAD & Design	CATIA, SolidWorks

Work Experience

Research Associate I Ottawa, Canada
University of Ottawa Nov. 2025 – Present

- Develop action/policy regularization methods to improve smoothness and stability of learned policies in continuous-control settings.
- Prototype and evaluate RL methods in GPU-accelerated simulation workflows using PyTorch and JAX (MuJoCo-based environments).
- Run controlled empirical studies to analyze temporal behavior, robustness, and reliability across changing conditions.
- Perform comprehensive literature reviews to inform research direction and identify gaps in current methodologies.
- Analyze experimental results and contribute to the peer-reviewed academic papers.

Research Assistant (Research Bursary) Ottawa, Canada
University of Ottawa & National Research Council of Canada (NRC) Sept. 2022 – Aug. 2025

- Designed and implemented the first Gymnasium-compatible RL environment in our project for wavefront sensorless adaptive optics (WSL-AO).
- Developed and validated model-free RL controllers for wavefront correction in satellite-to-ground optical communication scenarios.
- Developed action-regularized RL methods to reduce high-frequency oscillations and improve policy smoothness in complex environments.
- Transferred RL policies from simulation to a quadcopter platform via simulation-to-real transfer, targeting smooth and stable control.

Teaching Assistant (System Dynamics & Control Systems) Ottawa, Canada
University of Ottawa Jan. 2019 – Apr. 2022

- Supported undergraduate courses in system dynamics and control systems through tutorials, grading, and office hours.
- Assisted students with modeling, stability analysis, and feedback control design of dynamical systems using MATLAB/Simulink.
- Nominated for the Excellence in Teaching Assistant Award (2022).

Project Assistant (Robotics) Ankara, Turkey
Middle East Technical University & TÜBİTAK Nov. 2015 – Nov. 2017

- Contributed to an R&D project on a high-precision hybrid robotic deburring system involving a hexapod platform and an ABB manipulator.
- Modeled and analyzed human motor skills and motion primitives to support robotic process planning.
- Implemented and tested robotic motion and force-control strategies on experimental platforms.
- Contributed to conference publications related to robotic motion parametrization and skill transfer.

Education

PhD in Mechanical Engineering Ottawa, Canada
University of Ottawa Sept. 2018 – Oct. 2025

- Thesis: *Adaptive Policy Smoothing in Reinforcement Learning: Applications to Wavefront Sensorless Adaptive Optics and Robotics* [Link]
- Supervisors: Dr. Davide Spinello (uOttawa), Dr. Colin Bellinger (NRC, uOttawa), Dr. Ross Cheriton (NRC)

Master of Applied Science (Mechanical Engineering) Ankara, Turkey
Middle East Technical University Feb. 2015 – Feb. 2018

- Thesis: *Demonstration of Robotic Deburring Process Motor Skills from Motion Primitives of Human Skill Model* [Link]
- Supervisor: Dr. Erhan İlhan Konukseven

Selected Publications

Adaptive Policy Regularization for Smooth Control in Reinforcement Learning

Payam Parvizi, Abhishek Naik, Colin Bellinger, Ross Cheriton, Davide Spinello

Submitted to IEEE Transactions on Automation Science and Engineering, Preprint available at TechRxiv, 2026 [Link]

Action-Regularized Reinforcement Learning for Adaptive Optics in Optical Satellite Communication

Payam Parvizi, Colin Bellinger, Ross Cheriton, Abhishek Naik, Davide Spinello

In final production at Journal of the Optical Society of America B (Optica). Preprint available at Optica Open, 2025 [Link]

Reinforcement Learning Environment for Wavefront Sensorless Adaptive Optics in Single-Mode Fiber Coupled Optical Satellite Communications Downlinks

Payam Parvizi, Runnan Zou, Colin Bellinger, Ross Cheriton, Davide Spinello

Photonics, 2023 [Link]

Wavefront Sensorless Adaptive Optics for Free-space Satellite-to-Ground Communication using Reinforcement Learning

Runnan Zou, **Payam Parvizi**, Ross Cheriton, Colin Bellinger, Davide Spinello

Workshop on Communications and Observations through Atmospheric Turbulence (COAT), 2023 [Link]

Dynamic movement primitives and force feedback: Teleoperation in precision grinding process

Kemal Açıkgöz, **Payam Parvizi**, Abdulhamit Dönder, Musab Çağrı Uğurlu, Erhan İlhan Konukseven

IEEE International Conference on Electrical and Electronics Engineering (ELECO), 2017 [Link]

Robotic Deburring via Motion Primitives Parametrization of Robotic Deburring Process with Motor Skills from Motion Primitives of Human Skill Model

Payam Parvizi, Musab Çağrı Uğurlu, Kemal Açıkgöz, Erhan İlhan Konukseven

IEEE International Conference on Methods and Models in Automation and Robotics (MMAR), 2017 [Link]

Presentations

- *Poster Presentation* – **Photonics North (2025)**, Ottawa, Canada
- *Seminar Talk* – **MCG Seminars (2025)**, Department of Mechanical Engineering, University of Ottawa
- *Poster Presentation* – **Technology for the Digital Transformation of Society (2023)**, University of Ottawa
- *Seminar Talk* – **AI for Design Seminar (2023)**, National Research Council of Canada (NRC)
- *Poster Presentation* – **Photonics North (2023)**, Montreal, Canada

Certificates

MicroMasters in Statistics and Data Science

Massachusetts Institute of Technology (MITx)

2019-2020

- Machine Learning with Python
- Fundamentals of Statistics
- Probability